Learning Orientation Information From Frequency-Domain for Oriented Object Detection in Remote Sensing Images

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***Abstract*— Object detection in remote sensing images (RSIs) poses great difficulties due to arbitrary orientations, various scales, and dense location of the targets over the ground. Recent evidence suggests that encoding the orientation information is of great use for training an accurate object detector for oriented object detection (OOD). In this article, we propose a new frequency-domain orientation learning (FDOL) module with two main components: the frequency-domain feature extraction (FFE) network and an orientation enhanced self-attention layer (OES-Layer). The FFE network models the interactions among spatial locations in the frequency domain to determine the frequency of spatial features. Then, these features are fed into our OES-Layer to learn the orientation information. Moreover, the orientation weights are adopted to guide the feature selection in a self-attention (SA) architecture, using them as a control gate to emphasize the spatial responses of target instances. Considering that the original similarity weights (calculated by the SA algorithm) do not distinctly model the orientation variation, the considered orientation weights provide an efficient asset to emphasize the orientation of objects. Extensive experiments on the DOTA and HRSC2016 datasets demonstrate that our method achieves state-of-the-art performance among single-scale methods while achieving competitive performance over multiscale methods.**

***Index Terms*— Frequency analysis, orientation weight, oriented object detection (OOD), remote sensing images (RSIs), self-attention (SA).**

I. Introduction

OBJECT detection in remote sensing images (RSIs) plays a vital role in many fields such as traffic monitoring, environmental monitoring, and urban planning. Different from natural images, objects in aerial images often appear in oblique orientation and densely packed, which poses great challenges to delineate the objects using oriented bounding boxes (OBBs). Driven by the excellent performance of convolutional neural networks (CNNs), many methods have resorted to CNNs to detect or classify various instances [1], [2], [3], [4], [5] and advance the progress of object detection in RSIs.

Ordinary CNNs face great difficulties for representing instances with arbitrary orientation due to the inexplicit operation of modeling the orientation variation. Many recent methods have proposed a series of orientation information learning schemes to boost the performance of oriented object detection (OOD). Yang and Yan [1] summarized the different rotated rectangle boundary methods: five parameters and 90-angle regression, five parameters and 180-angle regression, and eight parameters regression, and proposed the circle smooth label (CSL) to solve the periodicity of the angle and decrease the training error between adjacent angles. They explored the orientation information in the annotation level and defined a new learning scheme for OOD. Other OOD methods heuristically defined [2], [6] or adaptively learned candidate proposals [7], [8] with different scales, angles, and aspect ratios. These are effective methods to explore the orientation information in RSIs. Objects with different angles and aspect ratios (e.g., small vehicles and ships) can be captured with the least background noise by adopting a densely anchors-based generation strategy. Although the orientation information is excavated by the densely anchors, these methods significantly increase the computational overheads. Methods [9], [10], [11] paid attention to extracting orientation-sensitive and orientation-invariant features to wrap the region features precisely. These approaches utilized additional channels to record the orientation information to achieve a better representation of oriented objects. Many recent studies [12], [13], [14], [15], [16] attempted to generate better features to improve the performance of OOD. The attention mechanism, as a most frequently used technique in feature selection, is widely utilized to emphasize the important spatial features and suppress redundant ones.

However, using a variety of convolution layers to define objects under different transformations (scales and angles) suffers from several CNN-related problems. Although the dense anchor generation strategy brings a valid way to capture oriented objects with different angles and scales, this algorithm requires more computations to obtain the oriented features,

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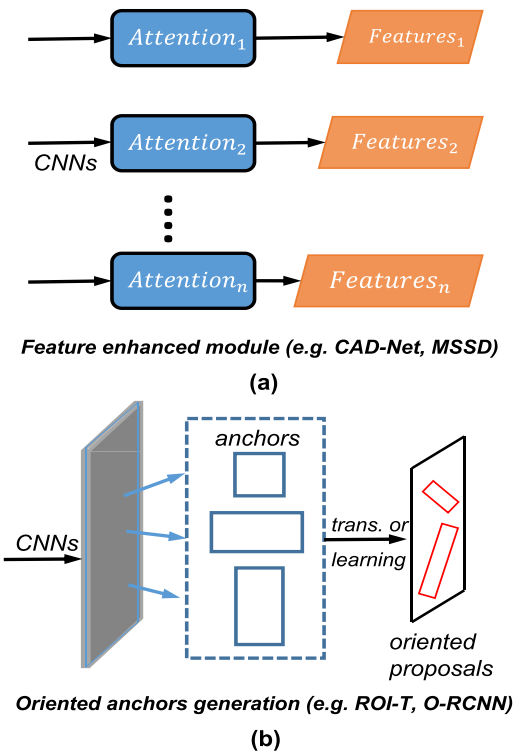


Fig. 1. Overview of recent orientation-based information extraction methods. (a) CNN-based feature enhanced module. (b) Oriented anchors generation strategy. The oriented proposals are generated by the transfer function or learning fashion.

which are highly inefficient. As shown in Fig. 1(a), the attention mechanism or other orientation information extraction components are constructed to emphasize the response of small objects in RSIs and, then, an OBB regression strategy is used to better locate the detected instances. However, features extracted from CNNs are limited by the properties of convolutional operations. Fig. 1(b) illustrates the process of generating the oriented proposals. After defining the horizontal anchors with different aspect ratios, recent methods utilized a transformation function or learning scheme to model the rotated proposals. However, these detectors require more parameters, which could cause massive computation and memory footprint to encode the orientation information. Recent works attempt to alleviate this issue by means of the label assignment or geometric computations but still suffer from significant computation costs.

In this article, we propose a new frequency-domain orientation learning (FDOL) network to detect oriented objects in RSIs. Frequency-domain analysis is an efficient tool to model the interactions of different instances in frequency space. After converting signals from the time domain to the frequency domain, all the high-frequency and low-frequency features are concatenated in the channel dimension and used as the auxiliary information to enhance the directional sensitivity of CNN features. First, a frequency-domain feature extraction (FFE) network is proposed to extract the spatial and frequency features with different scales and angles. Then, an orientation enhanced self-attention layer (OES-Layer) is constructed to calculate the orientation weights, which are adopted to model the different scale- and angle-level relationships among all objects. The orientation weights aim to supplement the orientation information for feature selection, which can be seamlessly integrated into the self-attention (SA) architecture. Finally, our FDOL unifies the proposed FFE network and OES-Layer to efficiently detect oriented objects in RSIs. The main innovative contributions of this article are threefold.

1) The orientation weights are adopted in our OES-Layer to assist the feature selection process. The interactions among spatial locations, obtained from a SA algorithm, are further refined by the constructed orientation weights.
2) We propose a new FFE network to extract orientation features with various scales and angles. Different from the orientation mining process conducted by CNNs, our FFE network provides an efficient tool to discriminate directional features.
3) After combining the FFE network and an OES-Layer with the backbone, our newly proposed FDOL network outperforms many state-of-the-art single-scale methods on DOTA and HRSC2016 datasets, which demonstrates the effectiveness of extracting orientation information from the frequency domain for OOD in RSIs.

The rest of this article is organized as follows. Section II reviews recent OOD methods and frequency domain learning approaches in CNNs. In Section III, we introduce the details of our method. The results obtained by our model and a comparison with many state-of-the-art approaches are given in Section IV. Finally, Section V concludes the article with some remarks and hints at plausible future research lines.

II. Related Works

A. Object Detection in RSIs

Compared with object detection in natural scenes, OOD in RSIs is much more challenging due to the arbitrary orientation and large aspect of objects. With the rapid development of object detectors in general scenes, many artworks [17], [18], [19], [20], [21], [22], [23] have been adopted for OOD in RSIs. Average precision (AP) [24] is the most frequently used evaluation metric in object detection. Recent works used mean average precision (mAP) to evaluate the performance of their methods. Current object detection methods identify objects using horizontal bounding boxes (HBBs) and can be divided into anchor-based [25] and anchor-free detectors [26] (e.g., Faster R-CNN and CenterNet). Based on the dense anchors generation strategy, the anchor-based methods pose great difficulties for tuning the network parameters. Objects with large shape variations are difficult to be detected via fixed scales and predefined aspect ratios of anchors. The anchor-free methods directly regress the bounding box without utilizing a set of preset anchors. However, they also meet difficulties when learning the dense locations of the oriented targets.

To achieve better detection performance, current works tend to model the orientation information to boost the directional discriminativity of CNN features. Ding *et al.* [7] proposed a RoI transformer to learn the rotated RoIs (RRoIs) from

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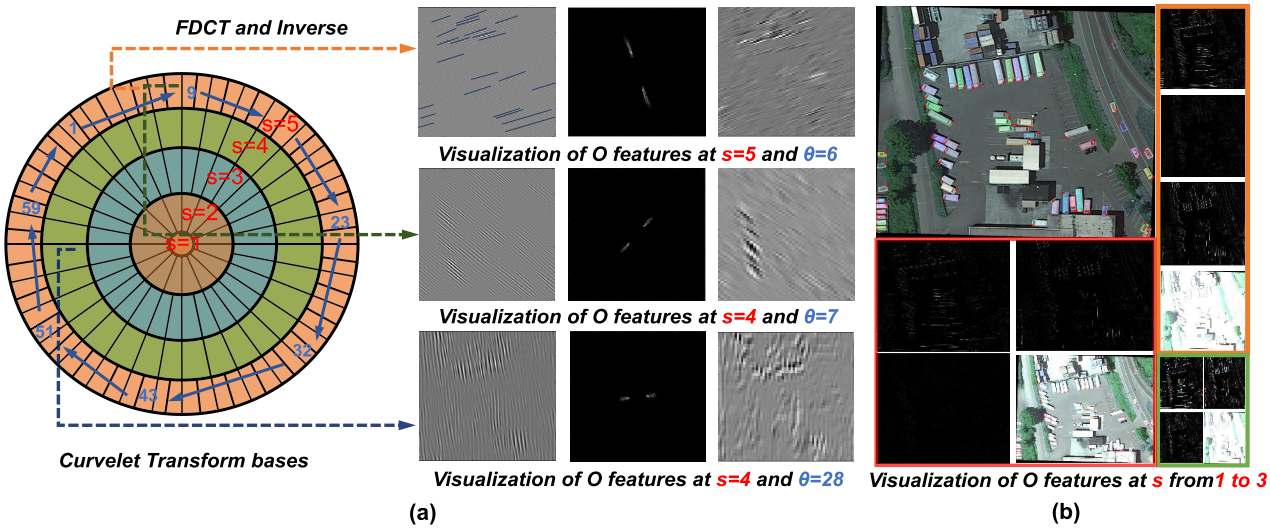


Fig. 2. Graphical illustration of: (a) curvelet transform (CT) and (b) wavelet transform (WT). We utilize multiple concentric circles instead of rectangles to represent the bases of the CT for better reading ("O features" represents orientation features).

horizontal RoIs (HRoIs). Ming *et al.* [27] proposed a sparse label assignment (SLA) strategy to alleviate the inconsistencies between classification and regression. Based on the posterior IoU of detection, the most high-quality sparse anchors can be selected to reduce the calculation burden. Many studies tried to utilize better features to improve the performance of detection [12], [13], [14], [15]. Zhang *et al.* [12] proposed the attention-enhanced features to merge the global and local contexts information for detecting oriented objects with different scales. FMSSD [13] aggregated context information to improve the accuracy of small object detection via the proposed unified feature-merged network. Fu *et al.* [15] constructed a feature-fusion architecture to reduce the fragment of adjacent instances and background areas caused by axis-aligned boxes.

Defining different forms of the OBBs to represent the oriented objects has been discussed in many recent works [1], [28], [29], [30]. These methods explored the use of orientation information at the annotation level. Yang *et al.* [28] proposed the densely coded labels to avoid the problem of out-of-bounds angles. They utilized fine-grained angle classification strategy to replace the angle regression, making the network converge easier. Qian *et al.* [29] constructed the modulated loss to better supervise the bounding box regression optimization to unify the boundary condition. Yang *et al.* [30] further adopted the Gaussian Wasserstein distance loss to address the inconsistencies between the localization accuracy and training loss caused by the oriented boundary problems.

B. Self Attention

As an important component in transformer structure, SA algorithm [31] has advanced many tasks [5], [32], [33], [34], [35] in RSIs. Hong *et al.* [5] proposed a novel transformer-based backbone network for hyperspectral image classification. In order to reduce the information dropping-out, a cross-layer skip connection strategy is constructed to adaptively fuse soft attention maps across layers. Zhang *et al.* [32] proposed spectral-spatial SA module to adaptively integrate long-range correlation over local features for hyperspectral image classification. However, due to the lack of the labeled data, it is urgent to develop unsupervised techniques for pixelwise task. A multiscale SA deep clustering-based octave convolution [33] network is constructed to extract multiscale features while reducing speckle noise. K-means++ algorithm is used to suppress the speckle noise without any supervision. Considering the computational costs rise significantly as the deeper layers are used to extract high-level semantic features, Sun *et al.* [34] proposed a spectral–spatial feature tokenization transformer (SSFTT) to efficiently capture deep semantic features. A combination of 3-D and 2-D convolution layer is introduced for feature transformation.

Algorithm 1 Pseudocode of FFE

```
# X:original images
# Extracted orientation features by CT
f = CT(X)
f_o = concatenate(f, 3)
# Extracted orientation features by WT
f = WT(X)
f_o = concatenate(f, 3)
```

C. Frequency Domain Learning

Frequency analysis is a fundamental powerful technique in the signal processing fields. Recent applications have witnessed success in introducing frequency analysis in deep learning fields. In [36], Ehrlich *et al.* utilized frequency analysis in conjunction with CNNs for JPEG encoding. Qin *et al.* [37] regarded the channel attention (CA) as a compression problem and generalized the CA in frequency domain with the discrete

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Algorithm 2 Pseudocode of Curvelet Transform (CT)

```
# X:original images
# s=6, θ=16
C = fdct_wrapping(X,0,2,6,16)
C = traversing_filtering_each(s, θ)
Y = ifdct_wrapping(C, 1)
# Fast Fourier Transform and Inverse
F = ifftshift(fft2(fftshift(Y)))
# Three results
spatial view = imagesc(real(Y))
frequency view = imagesc(abs(F))
coefficient view = reshape(imagesc(C))
```

Algorithm 3 Pseudocode of Wavelet Transform (WT)

```
# X:original images
# t=5
W = wavedec2(X,2,'Haar')
traversing all scale factor t
HVD_t = detcoef2('all',W,t)
CA_t = dappcoef2(W,'Haar',t)
# Four results
low_img = imagesc(CA_t)
horizontal coefficient = imagesc(H_t)
vertical coefficient = imagesc(V_t)
diagonal coefficient = imagesc(D_t)
```

cosine transform (DCT). Gueguen *et al.* [38] also trained CNNs on the block-wise DCT to expand the representation of pixels. In [39], a wavelet CNN is proposed to bring out spectral features for hyperspectral image (HSI) classification. The wavelet CNN is a variation of CNN for multiresolution learning, which reduces the cost of computing spectral features. Fnet [40] replaced the complicated SA layers with fast Fourier transform (FFT) algorithm, achieving nearly the same performance as the original SA operation, while the footprints of their method are also reduced. Similarly, GFNet [41] learned long-term spatial dependencies in the frequency domain. They utilized a pair of 2-D DFF and 2-D inverse Fourier transform (FT) to replace the SA algorithm. Wu *et al.* [42] for the first time proposed a rotation-invariant spatial-frequency descriptor for geospatial object detection by strict theoretical proof on the rotation-invariance, which is applicable to continuously arbitrary angle rotation of objects, achieving the significant detection performance improvement. Cao *et al.* [43] proposed a dual-frequency attention fusion network (DFAF-Net) to adaptively fuse the CNN features and frequency features to improve classification performance. The dual-frequency data were transformed by the proposed dual-frequency attention module and sent to the joint feature extraction (JFE) module to emphasize the representation of features. Method [44] used wavelet transform (WT) to detect edges in SAR Images. Different scales of features are transformed by WT to capture the edges information without any supervision.

III. Proposed Method

The main idea of the proposed FDOL is utilizing frequency analysis to extract the orientation information and boost the representativeness of features in CNNs. First, a FFE network is constructed to model the space and frequency information with different scales and angles. If we only consider the interactions among spatial locations achieved by the SA algorithm, it is not possible to distinctly model the orientation variations. The orientation weights are adopted to assist the feature selection process in the SA architecture. The proposed OES-Layer unifies the orientation weights with the SA algorithm and gathers features with a sliding convolution window in the frequency and space domains. We integrate the FDOL into a ResNet backbone network to boost the performance of OOD.

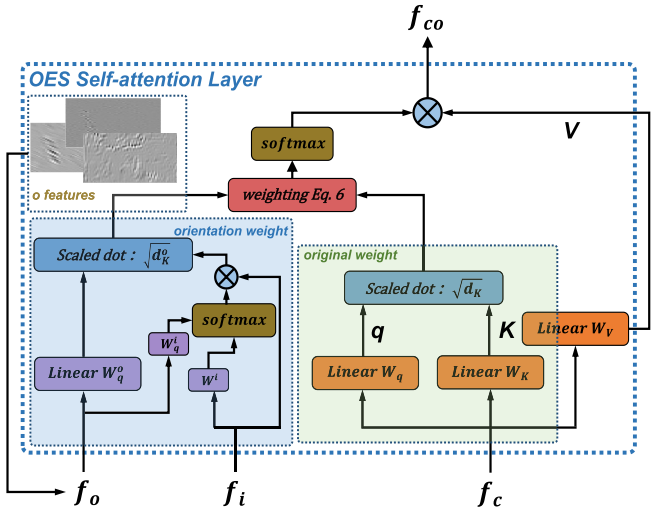


Fig. 3. Overview of our OES-Layer. Compared to the original similarity weight, we propose a new orientation weight which is defined by orientation features extracted from FFE.

A. Overall Framework

One image $I \in R^{1024\times1024}$ is taken as the input of ResNet and the extracted CNN features of every res-blocks have four different scales: $f_{c,1} \in R^{256\times256}$, $f_{c,2} \in R^{128\times128}$, $f_{c,3} \in R^{64\times64}$, $f_{c,4} \in R^{32\times32}$. Thus, our method aims to extract four different scales of orientation features: $f_{o,1} \in R^{256\times256}$, $f_{o,2} \in R^{128\times128}$, $f_{o,3} \in R^{64\times64}$, $f_{o,4} \in R^{32\times32}$. The orientation features, extracted from the proposed FFE network, are used to enhance the directional sensitivity of CNN features via our proposed OES-Layer. The overall framework of the proposed OES-layer can be summarized as

$$f_{c,s}^E = \text{OES} - \text{Layer}(f_{c,s}, f_{o,s}), \quad s = 1, 2, 3, 4 \tag{1}$$

where $f_{c,s}^E$ denotes the enhanced CNN features which are fed into the detector network for object detection

$$f_{o,s} = \text{FFE}(I, s), \quad s = 1, 2, 3, 4. \tag{2}$$

B. Frequency-Domain Feature Extraction (FFE) Network

We start by introducing the curvelet transform (CT), which plays a vital role in the field of digital signal processing and is an important component in our FDOL. The CT is

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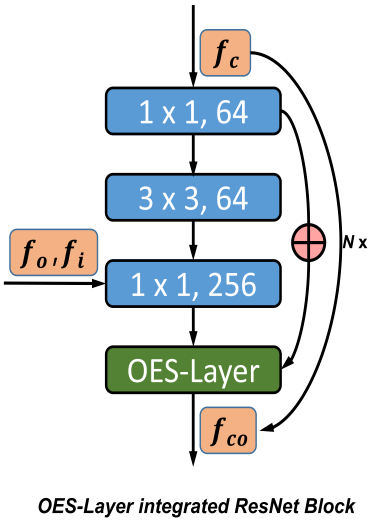


Fig. 4. Illustration of how our OES-Layer can be merged into a ResNet res-block. f_c, f_o, and f_i, respectively, denote CNN features extracted by the backbone network, orientation features extracted by our FFE, and initialized features. $N\times$ denotes the cycles of the res-block.

a multiscale geometric transformation which can effectively represent the edge curves of different angles. Similar to the 2-D FT, the 2-D CT essentially sets up a set of bases and resolves the correlation values of the spatial image on these bases, which constitutes the expression of the image on the CT basis

$$C(s,\theta,p_1,p_2)=\sum_{x\geq 0,y\geq 0}^{x\leq M,y\leq N} I[x,y]\cdot\phi_{s,\theta,p_1,p_2}[x,y] \tag{3}$$

where $I[x, y]$ denotes the input image and M, N represent the height and width of image. p_1, p_2 represents the spatial position of the CT. Different from 2-D FT bases, CT uses s and θ to depict various scales and angles of local curves. $\phi[x, y]$ denotes the bases calculated by the above parameters s, θ, p_1, p_2. Equation (3) can be reformulated as

$$\text{CT}=\text{IFFT}(\text{FFT}(C)\times\text{FFT}(I[x,y])) \tag{4}$$

where FFT and IFFT, respectively, denote the FT and the inverse FT. $I[x, y]$ denotes the input image. Divided by the frequency domain, a visualization of CT bases is illustrated in Fig. 2. The outer fan boxes represent more detailed edge features of small oriented objects. Considering that discrete signals can be easily processed by computers, we use the fast discrete CT (FDCT) to efficiently extract the frequency features in our FFE.

In practice, the minimum value of parameter s is set to 6 for CT. Thus, a total of six scales of frequency-domain features are extracted by CT. In addition, the orientation features extracted from CT are not the regular square shape. Thus, we only select the orientation features that are close to the size of the CNN features and resize these features to the standard scale

$$f_{o,s}=r_s(\text{CT-FFE}(I,s)),\quad s=1,2,3,4 \tag{5}$$

where r_s denotes the resize function.

Moreover, we also use the WT to extract orientation information with the Haar kernel $\text{Kernel}_{\text{high},s}$ and utilize a scaling function $\text{Kernel}_{\text{low},s}$ to control the scale transformation according to size of output from different residual block. Following [39], four kernels are used for WT as

$$f_{L,L}=\begin{bmatrix}1&1\\1&1\end{bmatrix}\quad f_{L,H}=\begin{bmatrix}-1&-1\\1&1\end{bmatrix}$$
$$f_{H,L}=\begin{bmatrix}-1&1\\-1&1\end{bmatrix}\quad f_{H,H}=\begin{bmatrix}1&-1\\-1&1\end{bmatrix}. \tag{6}$$

We resize the input image I from 1024×1024 to 512×512 and the scale parameter s is set to 4. Thus, a total of four scales of orientation features are extracted by WT: horizontal features $f_{h,s}|_{s=1}^4$, vertical features $f_{v,s}|_{s=1}^4$, diagonal features $f_{d,s}|_{s=1}^4$, low-resolution features $f_{LR,s}|_{s=1}^4$. That is

$$\begin{aligned}f_{o,s}&=\text{WT-FFE}(I,s)\\&=(f_{h,s},f_{v,s},f_{d,s},f_{\text{LR},s}),\quad s=1,2,3,4.\end{aligned} \tag{7}$$

Fig. 2(a) shows a visualization of the orientation features extracted by FDCT. When the scale s and angle θ are set to (5 and 6), (4 and 7), and (4 and 28), the spatial responses of target instances with different scales and angles are clearly depicted. Fig. 2(b) shows the orientation features extracted by WT. The horizontal, vertical, and diagonal edges of target instances with different scales are also distinctly visualized.

C. Orientation Enhanced Self-Attention Layer (OES-Layer)

1) Overall Architecture: Recent advances [37], [40], [41] demonstrate that methods based on the cooperation between frequency domain learning and CNNs achieve competitive performance in many tasks. Thus, our OES-Layer aims to extend the original SA similarity weight into two components: the original weight and a new orientation weight. The latter component utilizes frequency domain learning to model different scale- and angle-level relationships among all objects, boosting the orientation-sensitiveness of CNN features. Experimental results prove the importance of the proposed orientation weights.

The SA algorithm uses inner products to model the similarity weight between a query matrix and a set of key vectors. These results of inner products are normalized by the softmax function. It can be formulated as

$$\text{Attention}(q,K,V)=\text{softmax}\left(\frac{qK^T}{\sqrt{d_k}}\right)V \tag{8}$$

where q, K, and V represents query, keys, and values matrices, which are transformed by the corresponding linear functions: W_q, W_K, and W_V. The original similarity weight w_c is calculated as:

$$w_c=\frac{\text{dot}(W_K f_c, W_q f_c)}{\sqrt{d_k}} \tag{9}$$

where f_c represents the features extracted from CNNs and dot denotes the matrix multiplication. Equation (9) projects original features f_c into the sub-spaces to model how well they match to each other.

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TABLE I
TRAINING TIME (H), FPS (TASK/S), FLOPS (G), AND PARAMETERS (M) COMPARISONS ON BASELINE, SELF-ATTENTION ENHANCED AND OUR ORIENTATION WEIGHT ENHANCED SELF-ATTENTION METHODS IN THE DOTA-V1.0 DATASET

Methods	Param. (M)	FPS(tasks/s)	FLOPs(G)	Train. (h)
Baseline	31.65	7.8	134.08	23.5
SE	31.81	4.9	135.01	33.2
Ours (WT+O-SE)	32.16	4.4	138.70	39.2
Ours (CT+O-SE)	32.27	4.2	142.30	41.1

2) Orientation Weight: As illustrated in Fig. 3, orientation weight w_o is computed as

$$w_o = \frac{\mathrm{dot}\left(W_K^i f_i, W_q^o f_o\right)}{\sqrt{d_k^o}} \tag{10}$$

where W_q^o is linear function and plays a similar role as in (9). f_o represents the orientation features extracted from the FFE network. In our experiments, we set s to 6 and θ to 16 for the FDCT and set s to 4 for the WT. These orientation features are embedded into one high-level representation by a convolution layer. The f_i denotes an initialized key vector, aiming to reduce the memory footprints of our model. Moreover, we use an extra SA to boost the key vector features, and the (10) can be further formulated as

$$w_o = \frac{\mathrm{dot}\left(S - A(f_i, f_o), W_q^o f_o\right)}{\sqrt{d_k^o}} \tag{11}$$

where $S - A$ denotes a SA operation which is introduced to capture the global and local contextual information to expand the representation of the key matrix in the calculation of orientation weights. First, the orientation features f_o and the initialized key vectors f_i are fed into the SA algorithm. Second, we transfer f_o and f_i with two different linear layers W_q^i and W^i. Finally, in order to reduce the memory cost, the final enhanced key vectors are formulated as

$$S - A(f_i, f_o) = \sigma\left(\mathrm{dot}\left(W^i f_i, W_q^i f_o\right)\right) f_i \tag{12}$$

where σ denotes softmax function. Finally, the comprehensive weight w_{co} is formulated as:

$$w_{co} = \alpha_1 w_c + \alpha_2 w_o \tag{13}$$

where α_1 and α_2 represent weighting parameters which are set to 1.0 in our experiments. $+$ denotes the elementwise summation. With the guidance of w_{co}, our detection network learns to extract the orientation information to further boost the performance of object detection in RSIs.

3) Integration With ResBlock: The details of the exact position of our module within a ResBlock in ResNet [45] are shown in Fig. 4. Inspired by [46], we apply FDOL on the convolution outputs in each block to progressively propagate orientation information during network training. Moreover, our FDOL has the same input and output dimension, and can hence be regarded as a control gate to be plugged into any network architecture.

IV. RESULTS AND DISCUSSION

This section covers the experimental settings and evaluation results for our newly proposed FDOL. All our experiments are based on Pytorch and MATLAB tools, implemented on two blocks of 11 GB memory NVIDIA GTX 1080Ti GPUs, and evaluated on the HRSC2016 [57] and DOTA [47] datasets.

A. Datasets

1) DOTA: DOTA contains 2806 aerial images with scale range from 800 × 800 to 4000 × 4000 and 15 categories of 188 282 examples are annotated in DOTA-v1.0, 16 categories of 402 089 instances are annotated in DOTA-v1.5. Compared with DOTA-v1.0, DOTA-v1.5 contains the same categories: plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC) and has a new category: container crane (CC). Following the settings in previous methods [11], we use both training and validation sets for training. The rest 2/6 of original images are used as testing set.

2) HRSC2016: HRSC2016 is devoted to ship detection in RSIs. It contains 1061 images, including 436 images for training, 181 samples for validation and 444 images for testing. All the objects are annotated with OBB. Following the standard protocol, both training and validation sets are all used for training and the rest of the test set for testing. The images are resized to 800 × 512 for testing. Only horizontal flipping is applied during training.

B. Implementation Details and Evaluation Criteria

We adopt [10] as our baseline method, which combined ReResNet and ReFPN networks. For fair comparison, we also use the same settings as [10]: the location of each pyramid stage set 15 anchors and samples 512 RoIs with 1:3 positive to negative ratio for training. We use the SDG optimizer and the momentum and weight decay are set to 0.9 and 0.0001. The learning rate is set to 0.005. For the DOTA dataset, the original images are cropped into 1024 × 1024 patches with overlap of 200 between two adjacent patches. Our models are trained using 12 epochs with learning rate divided by 10 at epoch 8 and epoch 11. For the HRSC2016 dataset, we hold the original aspect ratios of each image, and resize the shorter sides to 800. The longer ones do not exceed 1333. We train our models in 36 epochs with learning rate divided by 10 at epoch 24 and epoch 33. We use two GPUs and the batch size is set to 4 for training.

AP [24] is the most widely used evaluation criteria in the object detection tasks. Following the standard protocol, we use mAP to measure the performance of our FDOL on the DOTA dataset. For the HRSC2016 dataset, the mAP (07) and mAP (12) denote the mAP metrics are calculated under Pascal VOC 2007 and 2012 criteria [24]. The mAP (0.5:0.95) represents the mAP is calculated over different IoU thresholds from 0.5 to 0.95.

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TABLE II

QUANTITATIVE RESULTS OF DIFFERENT OBJECT DETECTION MODELS ON DOTA-V1.0. RESULTS OF METHODS ARE PARTLY OBTAINED FROM THEIR RELEASED CODE AND RECONSTRUCTED WHEN NEEDED

Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
single-scale:																	
FR-O [47]	R101	79.42	77.13	17.70	64.05	35.30	38.02	37.16	89.41	69.64	59.28	50.30	52.91	47.89	47.40	46.30	54.13
ICN [48]	R101	81.36	74.30	47.70	70.32	64.89	67.82	69.98	90.76	79.06	78.20	53.64	62.90	67.02	64.17	50.23	68.16
CADNet [12]	R101	87.80	82.40	49.40	73.50	71.10	63.50	76.60	90.90	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
RoI Transformer [7]	R101	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
DRN [49]	H-104	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
CenterMap [50]	R50	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.61	49.36	66.19	72.10	72.36	58.7	71.74
SCRDet [51]	R101	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
FAOD [52]	R101	90.21	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	74.17	69.69	64.86	73.28
R^3Det [11]	R152	89.49	81.17	50.53	66.10	70.92	78.66	78.21	90.81	85.26	84.23	61.81	63.77	68.16	69.83	67.17	73.74
S^2A-Net [9]	R50	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
Oriented R-CNN [53]	R50	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87
ReDet [10]	ReR50	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
Oriented R-CNN [53]	R101	88.86	83.48	55.27	76.92	74.27	82.10	87.52	90.90	85.56	85.33	65.51	66.82	74.36	70.15	57.28	76.28
FDOL (Ours)	ReR50	88.82	84.15	56.25	75.04	74.89	83.88	88.14	90.91	87.97	85.42	59.44	66.36	77.05	74.11	65.89	**77.22**
multi-scale:																	
FR-O [47]	R101	88.44	73.06	44.86	59.09	73.25	71.49	77.11	90.84	78.94	83.90	48.59	62.95	62.18	64.91	56.18	69.05
RoI Transformer [7]	R101	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
DRN [49]	H-104	89.71	82.34	47.22	64.10	76.22	74.43	85.84	90.57	86.18	84.89	57.65	61.93	69.30	69.63	58.48	73.23
FAOD [52]	R101	90.21	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	74.17	69.69	64.86	73.28
Gliding Vertex [6]	R101	89.64	85.00	52.26	77.34	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
CenterMap [50]	R101	89.83	84.41	54.60	70.25	77.66	78.32	87.19	90.66	84.89	85.27	56.46	69.23	74.13	71.56	66.06	76.03
OWSR [54]	R101	90.41	85.21	55.00	78.27	76.19	72.19	82.14	90.70	87.22	86.87	66.62	68.43	75.43	72.70	57.99	76.36
S^2A-Net [9]	R50	88.89	83.60	57.74	81.95	79.94	83.19	89.11	90.78	84.87	87.81	70.30	68.25	78.30	77.01	69.58	79.42
ReDet [10]	ReR50	88.81	82.48	60.83	80.82	78.34	86.06	88.31	90.87	88.77	87.03	68.65	66.90	79.26	79.71	74.67	80.10
GWD [30]	R152	89.66	84.99	59.26	82.19	78.97	84.83	87.70	90.21	86.54	86.85	73.47	67.77	76.92	79.22	74.92	80.23
EDA [55]	ReR50	89.92	83.84	59.65	79.88	80.11	87.96	88.17	90.31	88.93	88.46	68.93	65.94	78.04	79.69	75.78	80.37
FDOL (Ours)	ReR50	88.90	84.57	60.73	80.83	78.42	85.82	88.33	90.90	88.28	86.93	71.44	67.13	79.00	80.35	74.59	**80.41**

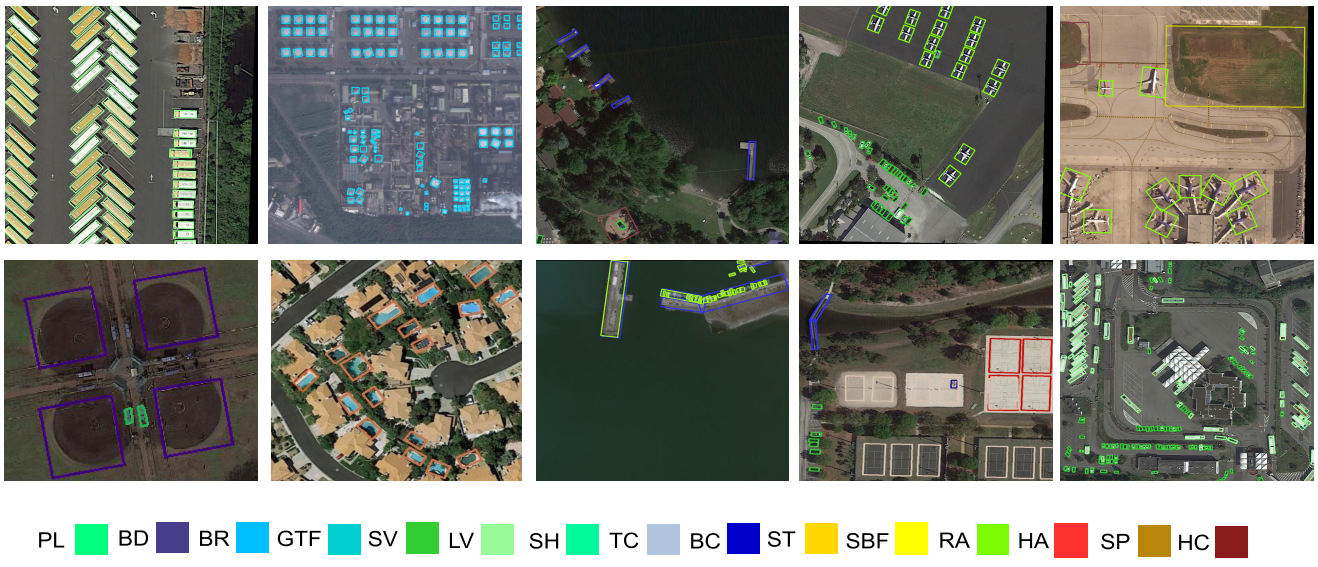


Fig. 5. Visualization of the detection results of our FDOL in the DOTA dataset.

C. Comparisons With State-of-the-Art

1) Experiments on DOTA: We conduct experiments for OBB tasks on DOTA-v1.0 and DOTA-v1.5 datasets. The training times, fps, FLOPs, and parameters of SA enhanced method and our FDOL are listed in Table I. All the methods are evaluated on one 1080ti GPU and the size of input image is set to 1024 × 1024. As shown in Table I, the baseline method

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TABLE III

QUANTITATIVE RESULTS OF DIFFERENT OBJECT DETECTION MODELS ON DOTA-V1.5. RESULTS OF METHODS ARE PARTLY OBTAINED FROM THEIR RELEASED CODE AND RECONSTRUCTED WHEN NEEDED

Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	mAP
RetinaNet-O [56]	–	71.43	77.64	42.12	64.65	44.53	56.79	73.31	90.84	76.02	59.96	46.95	69.24	59.65	64.52	48.06	0.83	59.16
FR-O [47]	R101	71.89	74.47	44.45	59.87	51.28	68.98	79.37	90.78	77.38	67.50	47.75	69.72	61.22	65.28	60.47	1.54	62.00
ReDet [10]	ReR50	79.20	82.81	51.92	71.41	52.38	75.73	80.92	90.83	75.81	68.64	49.29	72.03	73.36	70.55	63.33	11.53	66.86
FDOL (Ours)	ReR50	79.95	83.10	53.13	66.73	54.74	76.75	88.36	90.80	82.31	72.81	50.74	71.24	74.72	71.15	60.54	0.91	**67.88**
AICyber [54]	R101	88.4	85.4	56.7	74.4	63.9	72.7	87.9	90.9	86.3	85.0	68.9	76.0	74.1	72.9	73.4	37.9	74.70
CZH [54]	–	89.0	83.2	54.5	73.8	72.6	80.3	89.3	90.8	84.4	85.0	68.7	75.3	74.2	74.4	73.4	42.1	75.70
OWSR [54]	R101	88.2	86.4	59.4	80.0	68.1	75.6	87.2	90.9	85.3	84.1	73.8	77.5	76.4	73.7	69.5	49.6	76.60
ReDet [10]	ReR50	88.51	86.45	61.23	81.20	67.60	83.65	90.00	90.86	84.30	75.33	71.49	72.06	78.32	74.73	76.10	46.98	76.80
FDOL* (Ours)	ReR50	88.41	86.30	61.25	82.30	68.00	84.12	89.95	90.83	84.31	76.81	70.74	73.24	78.72	73.15	75.54	16.23	75.62
FDOL*‡ (Ours)	ReR50	88.64	85.48	63.25	82.87	67.58	82.98	90.91	90.16	85.61	77.23	70.14	74.05	79.25	73.66	76.85	45.26	**77.12**



Fig. 6. Visualization of the detection results of baseline methods and our FDOL in the HRSC2016 dataset. The red rectangle boxes cover the target ships which are miss-detected by the orange rectangle boxes.

TABLE IV

RESULTS OF DIFFERENT METHODS IN THE HRSC2016 DATASET

Methods	Backbone	mAP (07)	mAP (12)
RC2 [58]	–	75.70	–
RRPN [2]	R101	79.08	85.64
RoI Transformer [7]	R101	86.20	–
Gliding Vertex [6]	R101	88.20	–
PIoU [59]	DLA-34	89.20	–
DRN [49]	H-34	–	92.70
R^3Det [11]	R101	89.26	96.01
DAL [60]	R101	89.77	–
S^2A-Net [9]	R101	90.17	95.01
Oriented R-CNN [53]	R50	90.40	96.50
ReDet [10]	ReR50	90.46	97.63
Oriented R-CNN [53]	R101	90.50	97.60
FDOL (Ours)	ReR50	**90.64**	**97.93**

requires 23.5 h of training while has the fewest number of parameters and the fastest testing speed. The original self-attention (SE) enhanced method preserves a similar amount of parameters with the baseline (31.81 M versus 31.65 M). Moreover, as shown in Table VI, the SE enhanced method shows 0.65 mAP improvements with 41.3% extra training time (33.2 h). Similarly, the fps and FLOPs of SE enhanced method are also inferior to the baseline network (4.9 tasks/s versus 7.8 tasks/s and 135.01 G versus 134.08 G). Compared with the SE enhanced method, our WT+O-SE network achieves 0.32 mAP improvements with only 18% extra training time. Our CT+O-SE method achieves 0.14 mAP improvements with only 24% extra training time. Moreover, our WT+O-SE method has the similar parameters, fps and FLOPs with the SE enhanced method (32.16 M versus 31.81 M, 4.4 tasks/s versus 4.9 tasks/s and 138.70 G versus 135.01 G). Importantly, our WT+O-SE method is always superior to the baseline and SE-enhanced methods which demonstrates the effectiveness of excavating orientation information from frequency-domain for OOD in RSIs.

As shown in Table II, we list the results obtained by our proposed method and other existing methods on the DOTA-v1.0 dataset. All our results are evaluated on DOTA Evaluation server in a fixed format. In the single-scale training and testing strategy, our method achieves the best performance, which demonstrates the effectiveness of the proposed FDOL for OOD in RSIs. Compared with the best competitor (oriented R-CNN [53]), our method achieves 77.22 OBB mAP which is 0.94% higher. After the multiscale training and testing strategy are adopted, our method achieves 80.41 OBB mAP, which

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TABLE V
QUANTITATIVE RESULTS OF WT-FFE_ENHANCED AND CT-FFE_ENHANCED FDOL ON DOTA-V1.0 DATASET

Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
Baseline	ReR50	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
CT-FFE	ReR50	88.20	82.81	55.83	76.83	77.83	84.54	87.88	90.87	88.49	85.39	62.11	65.76	77.24	71.06	60.69	77.04
WT-FFE	ReR50	88.82	84.15	56.25	75.04	74.89	83.88	88.14	90.91	87.97	85.42	59.44	66.36	77.05	74.11	65.89	**77.22**

TABLE VI
EFFECTIVENESS OF OUR ORIENTATION WEIGHT ENHANCED SELF-ATTENTION IN THE DOTA-V1.0 DATASET

Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
Baseline	ReR50	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
SE	ReR50	88.86	84.32	55.41	78.63	77.56	83.72	87.96	90.83	87.45	85.10	61.61	63.85	76.81	71.44	60.00	76.90
O-SE	ReR50	88.82	84.15	56.25	75.04	74.89	83.88	88.14	90.91	87.97	85.42	59.44	66.36	77.05	74.11	65.89	**77.22**

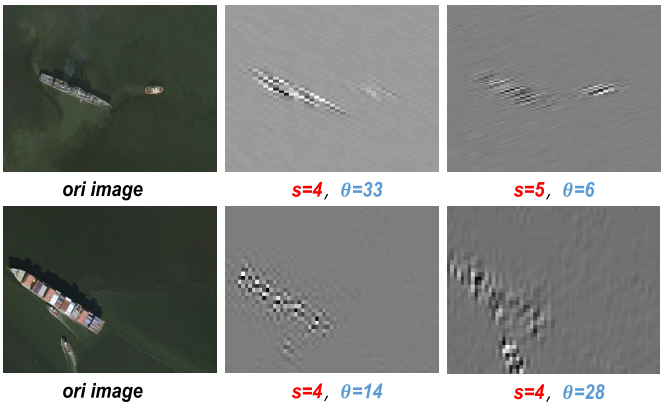


Fig. 7. Visualization of the orientation information on the HRSC2016 dataset.

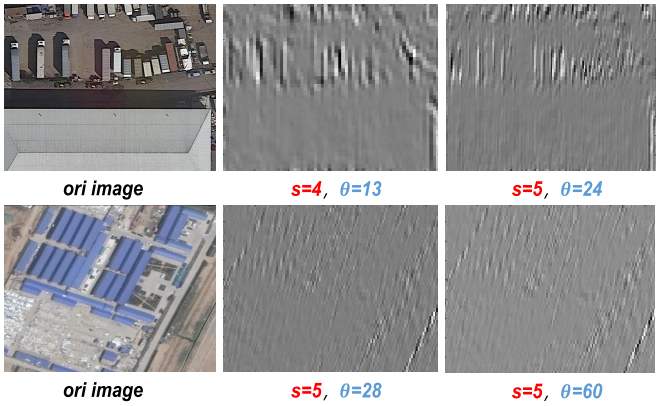


Fig. 8. Visualization of the orientation information on the DOTA dataset.

demonstrates the competitive performance compared to the other state-of-the-art methods. The detection results of our FDOL in the DOTA dataset are shown in Fig. 5. The proposed FDOL accurate detects different scale of oriented objects in RSIs.

As illustrated in Table III, we compare our FDOL with methods [10], [47], [54], [56] on the DOTA-v1.5 dataset. The symbol † means the results reported in [54]. The symbol $*$ denotes the multiscale training strategy. Our FDOL obtains 67.88 mAP and 1.01 better than ReDet [10] in single-scale training and testing. Additionally, our multiscale model achieves 75.62 mAP, which is relatively lower than the one obtained by ReDet. Specifically, the detection accuracy of CC is extremely small and unstable, ranging from 5.3 to 20.0, which is a major factor in our lower performance. Compared with other category samples, there are few number of category CC samples in the DOTA-v1.5 dataset, which demonstrates that our orientation weight needs a certain number of training samples. This will be addressed in our future work. FDOL*‡ denotes an ensemble of our method and the baseline network, which is also used in [54]. Thus, we also compare our proposed FDOL with the ensemble learning enhanced method [54]. Through feature compensation, our method achieves competitive performance. Most importantly, our single-scale model still achieves state-of-the-art performance.

2) Experiments on HRSC2016: Qualitative comparisons between our FDOL and the baseline method are visualized in Fig. 6. It can be clearly seen that our method achieves more accurate oriented bounding boxes. Table IV lists the mAP results of different methods on the HRSC2016 dataset. Our method achieves state-of-the-art performance with mAP of 90.64 and 97.93 under PASCAL VOC 2007 and 2012 metrics which demonstrates the effectiveness of excavating orientation information for OOD in RSIs. Figs. 7 and 8 show the visualization of the orientation information in the DOTA and HRSC2016 datasets. Our method effectively captures the edges of different oriented objects.

TABLE VII
QUANTITATIVE RESULTS OF WT-FFE_ENHANCED AND CT-FFE_ENHANCED FDOL IN THE HRSC2016 DATASET

Methods	mAP(75)	mAP(.5:.95)	mAP(07)	mAP(12)
Baseline	89.46	70.41	90.46	97.63
WT-FFE	89.44 (-0.02)	**71.76 (+1.35)**	**90.64 (+0.18)**	97.93 (+0.30)
CT-FFE	**89.71 (+0.25)**	71.69 (+1.27)	90.63 (+0.17)	**98.01 (+0.38)**

D. Ablation Studies

We conduct ablation experiments on the HRSC2016 and DOTA-v1.0 datasets to confirm the effectiveness of our proposed method. All the components are re-implemented with a rotation-equivariant network based on e2cnn [17] to better merge with the ReR50 backbone network.

1) Evaluation of FFE: The results of our CT and WT integrated FFE network are listed in Tables V and VII. Specifically, we use MATLAB tools to obtain the orientation information for the CT. For HRSC2016 dataset, our proposed

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TABLE VIII
EFFECTIVENESS OF OUR ORIENTATION WEIGHT ENHANCED SELF-ATTENTION IN THE HRSC2016 DATASET

Methods	mAP(75)	mAP(.5:.95)	mAP(07)	mAP(12)
Baseline	89.46	70.41	90.46	97.63
SE	**89.91 (+0.45)**	71.36 (+0.95)	90.56 (+0.10)	97.83 (+0.20)
O-SE	89.44 (-0.02)	**71.76 (+1.35)**	**90.64 (+0.18)**	**97.93 (+0.30)**

WT-FFE network achieves 90.64 and 97.93 mAP, which are 0.18 and 0.30 higher than the baseline method with VOC 2007 and VOC 2012 evaluation matrices. Moreover, due to stronger orientation feature extraction capability, our CT-FFE network achieves 90.63 and 98.01 mAP which are 0.17 and 0.38 higher than the baseline method with VOC 2007 and VOC 2012 evaluation metrics. For DOTA-v1.0 dataset, the proposed CT-FFE obtains 77.04 mAP which is 0.79 higher than the baseline method. In addition, our WT-FFE achieves 77.22 mAP which is 0.97 and 0.18 higher than the baseline method and our CT-FFE integrated model. All the experiments show the effectiveness of extracting orientation information to enhance the directional sensitivity of CNN features.

2) Evaluation of Orientation Weights: Here, we conduct experiments to verify the importance of the proposed orientation weights. First, we use original SA algorithm, which only explores the global spatial information to boost the performance of the baseline method. As listed in Tables VIII and VI, we denote the model with SA algorithm and our orientation weight_integrated SA algorithm as "SE" and "O-SE." For the HRSC2016 dataset, after merging SE algorithm into the baseline method, we achieve 90.56 and 97.83 mAP with VOC 2007 and VOC 2012 evaluation matrices which is 0.10 and 0.20 higher than baseline method. Then, we integrate the proposed O-SE to boost the performance of oriented ship detection in the HRSC2016 dataset. Our O-SE achieves the best 90.64 and 97.93 mAP, which is 0.18 and 0.30 higher than the baseline method. The interactions among spatial locations, obtained from the SE algorithm, are further refined by the constructed orientation weights. Moreover, we also conduct experiments on the DOTA-v1.0 dataset: the SE integrated model achieves 76.90 mAP which is 0.75 higher than the baseline method. The best performance is obtained by our O-SE integrated model, which is 0.97 and 0.32 higher than the baseline method and the SE integrated model. That is, our orientation weights could effectively capture orientated features to better represent the oriented instances in RSIs.

V. CONCLUSION AND FUTURE LINES

Orientation information is an important property, which could better represent oriented objects in RSIs. In this article, we propose a new FDOL module to better capture features with different scales and angles. A new FFE network is constructed to enhance the directional sensitivity of CNN features. Moreover, we propose an orientation weight which replaces the original similarity weight in the SA algorithm. The proposed orientation weights distinctly model the spatial correspondence of oriented objects and can be seamlessly merged into the SA algorithm. The performance of the resulting OES layer is superior to the original self-attention mechanism in OOD. Our FDOL has the same input and output dimensionality, and hence can be used in many popular network architectures. In the future, we will consider merging our FDOL with other networks in order to perform OOD.

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